Computational Finance Set 1

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1 Implied volatility computation

An Excel file contains European call and put option quotes.
Assume:

- Risk-free rate $r = 0.11\%$
- Act-365 day count convention
- No dividends

1. **Compute implied volatilities using the Bisection method.**
   For each market price C_{mkt}, solve:
   $$C_{\text{BS}}(S_0, K, r, T, \sigma) = C_{\text{mkt}}$$
   Plot implied volatility versus strike.
   (1.25 points)

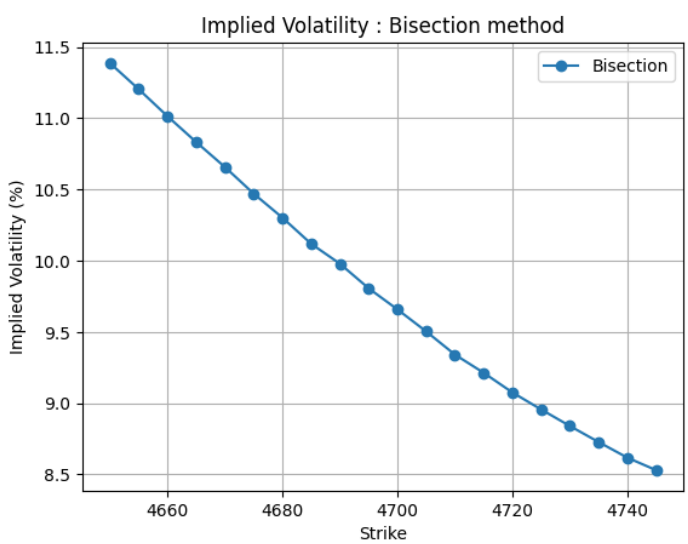

Figure 1: Implied volatility vs Strike for bisection method.

The implied volatility $\sigma^\star$ is computed by solving $C_{BS}(S_0, K, r, T, \sigma^\star) = C^{mkt}$ using the Bisection method on the interval $[10^{-6}, 5.0]$ with tolerance $tol = 10^{-6}$ and 200 iterations maximum.

2. **Compute implied volatilities using Newton's method:** Add these to the same plot.
(1.25 points)

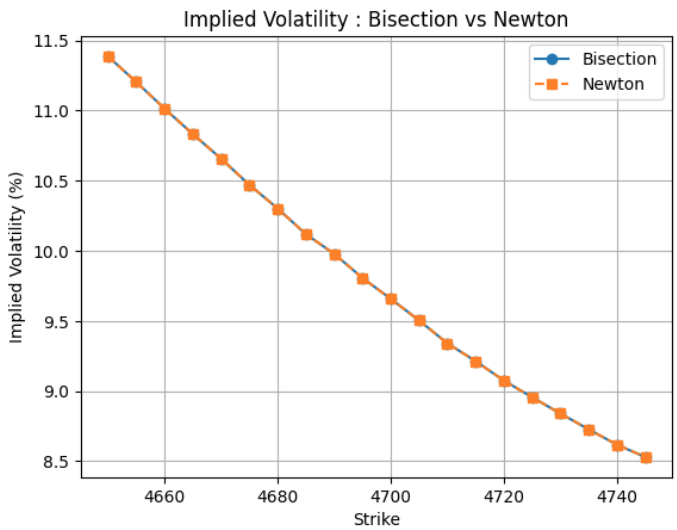

Figure 2: Implied volatility vs Strike for bisection and Newton methods.

The same equation is solved using Newton–Raphson, with update $\sigma_{n+1} = \sigma_n - (C_{BS}(\sigma_n) - C^{mkt})/\mathcal{V}(\sigma_n)$, where $\mathcal{V} = S_0\sqrt{T}\phi(d_1)$ is the Vega, and initial guess computed as in the slides of Lecture 3 which guarantees global convergence. We have used Newton method with tolerance $tol = 10^{-6}$ and 100 iterations maximum. Note that implied volatilities for both methods overlap, as expected since they solve the same equation to high precision.

3. Discuss possible reasons for differences with the IV column in Excel (numerical tolerance, initial guess, convergence issues, rounding, bid–ask spreads).
(0.5 points)

As seen in Figure 3, it can be observed that the implied volatility obtained by Bisection and Newton methods has small differences with the IV column in Excel.

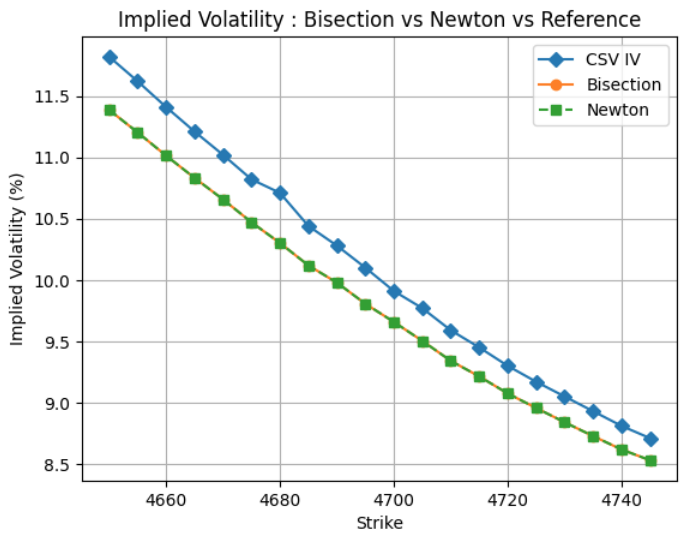

Figure 3: Comparison between IV from Excel and the ones obtained from Bisection and Newton methods.

The bid-ask spreads can be one possibility to explain this discrepancy, as in our case, we selected the midpoint between the bid and the ask as our C_{mkt} value. However, if the Excel uses the ask or bid prices, then, the IVs will differ.

Another possibility to explain the differences is the assumptions that we made. We supposed that there were no dividends and the price of the call was calculated under the Black-Scholes model, which assumes that trades are continuous, but in reality they are discrete. The Excel file had call and put option quotes with the ticker SPX which corresponds to the S&P 500 Index, in which there are companies that hand out dividends.

Also, rounding can be another fact to explain the difference. Quoted prices rounded to two decimals introduce IV errors of order $\Delta C/\mathcal{V}$, where ΔC is the rounding error. Let's deduce it: implied volatility is defined as $C_{BS}(\sigma^\star) = C_{mkt}$. If the market price is rounded, this means that the price that we observe is $\tilde{C_{mkt}} = C_{mkt} + \Delta C$, so, we are solving $\tilde{C_{BS}}(\sigma^\star) = C_{mkt}(\sigma^\star) + \Delta C$. By a first-order Taylor expansion of C_{BS} around the true $\sigma^\star$:

$$C_{BS}(\tilde{\sigma}^\star) \approx C_{BS}(\sigma^\star) + \frac{\partial C_{BS}(\sigma^\star)}{\partial \sigma} \cdot (\tilde{\sigma}^\star - \sigma^\star)$$

$$C^{mkt} + \Delta C \approx C^{mkt} + \mathcal{V}(\sigma^\star) \cdot (\tilde{\sigma}^\star - \sigma^\star).$$

Solving for the IV error, we get

$$\tilde{\sigma}^\star - \sigma^\star \approx \frac{\Delta C}{\mathcal{V}(\sigma^\star)}.$$

2 Risk-neutral pricing and change of numéraire

Assume the Black-Scholes model:

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}, \quad dB_t = rB_t dt.$$

(a) Derive:

$$V_0 = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[(S_T - K)^+\right].$$

We know that an European call has payoff $V_T = (S_T - K)^+$. In order to derive the price for V_0, we recall the Feynman–Kac formula, which establishes a link between parabolic PDEs and diffusion processes. So, if we consider the PDE

$$\frac{\partial V}{\partial t} + b(x,t)\frac{\partial V}{\partial x} + \frac{1}{2}\sigma^2(x,t)\frac{\partial^2 V}{\partial x^2} - r(x,t)V = 0,$$

with terminal condition $V(x,T) = g(x)$, and we consider X_s that solves the diffusion process

$$dX_s = b(X_s, s)\, ds + \sigma(X_s, s)\, dW_s, \quad s \geq t,$$

with initial condition $X_t = x$, then, the Feynman-Kac theorem states that

$$V(x,t) = \mathbb{E}\left[e^{-\int_t^T r(X_u,u)\,du} g(X_T) \mid X_t = x\right].$$

Applying it to our diffusion process S_t, and taking into account that $V(x,T) = (S_T - K)^+$, $r(x,t) = r$, $b(x,t) = rS_t$ and $\sigma(x,t) = \sigma$, then

$$\begin{aligned} V_0 = V(x,0) &= \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_0^T r\,du} V(x,T)\right] \\ &= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[(S_T - K)^+\right]. \end{aligned}$$

(b) Choose S_t as numéraire and define:

$$\frac{d\mathbb{Q}^S}{d\mathbb{Q}} = \frac{S_T/B_T}{S_0/B_0}.$$

Derive:

- The Radon-Nikodym derivative.
- The Brownian motion under $\mathbb{Q}^S$.

We choose S_t as the numéraire under the new measure $\mathbb{Q}^S$. Then, we rewrite $d\mathbb{Q}^s/d\mathbb{Q}$ as

$$\begin{aligned} \frac{d\mathbb{Q}^S}{d\mathbb{Q}} = \frac{S_T/B_T}{S_0/B_0} &= \frac{e^{-rT} S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right)T + \sigma W_T^{\mathbb{Q}}\right)}{S_0} \\ &= \exp\left(\sigma W_T^{\mathbb{Q}} - \frac{\sigma^2 T}{2}\right) = \exp\left(\int_0^T \sigma dW_s - \frac{1}{2}\int_0^T \sigma^2 ds\right), \end{aligned}$$

where we have used that in the Black-Scholes model $S_t = S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right) t + \sigma W_t^{\mathbb{Q}}\right)$. Then, the Radon-Nikodym derivative can be written as

$$\frac{d\mathbb{Q}^S}{d\mathbb{Q}} = \exp\left(\sigma W_T^{\mathbb{Q}} - \frac{\sigma^2 T}{2}\right).$$

Now, applying Girsanov's theorem with $\theta(t) = \sigma$, we get that the new Brownian motion under $\mathbb{Q}^S$ is

$$W_t^{\mathbb{Q}^S} = W_t^{\mathbb{Q}} - \sigma t. \tag{1}$$

Deriving we obtain

$$dW_t^{\mathbb{Q}^S} = dW_t^{\mathbb{Q}} - \sigma dt. \tag{2}$$

(c) Determine the new drifts under $\mathbb{Q}^S$.

Substituting (2) in the original SDE for the asset S_t, we obtain

$$\begin{aligned} dS_t &= rS_t dt + \sigma S_t \left(dW_t^{\mathbb{Q}^S} + \sigma dt\right) \\ &= \left(r + \sigma^2\right) S_t dt + \sigma S_t dW_t^{\mathbb{Q}^S}. \end{aligned}$$

Then, we observe that the new drift under $\mathbb{Q}^S$ is $\left(r + \sigma^2\right) S_t$ for the SDE of the asset S_t. Notice that the bank account B_t does not change because it has no Brownian component, consequently, the drift remains as rB_t. Observe that we can write under $\mathbb{Q}^S$

$$dS_t = \left(r + \sigma^2\right) S_t dt + \sigma S_t dW_t^{\mathbb{Q}^S}, \quad dB_t = rB_t dt.$$

(d) Recover the Black-Scholes formula using this approach.

To recover the Black-Scholes formula we split the expression found for V_0

$$\begin{aligned} V_0 &= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[(S_T - K)^+\right] \\ &= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[(S_T - K)\mathbf{1}_{S_t > K}\right] \\ &= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[S_T \mathbf{1}_{S_t > K}\right] - Ke^{-rT}\mathbb{E}^{\mathbb{Q}}\left[\mathbf{1}_{S_t > K}\right] \\ &= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[S_T \mathbf{1}_{S_t > K}\right] - Ke^{-rT}\mathbb{Q}\left(S_t > K\right). \end{aligned}$$

Now, we rewrite the first term using the change of measure defined in (b) by the Radon-Nikodym derivative:

$$\begin{aligned} e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[S_T \mathbf{1}_{S_t > K}\right] &= B_0/B_T \mathbb{E}^{\mathbb{Q}}\left[S_T \mathbf{1}_{S_t > K}\right] = S_0 \mathbb{E}^{\mathbb{Q}}\left[\frac{S_T/B_T}{S_0/B_0}\mathbf{1}_{S_t > K}\right] \\ &= S_0 \mathbb{E}^{\mathbb{Q}}\left[\frac{d\mathbb{Q}^S}{d\mathbb{Q}}\mathbf{1}_{S_t > K}\right] = S_0 \mathbb{E}^{\mathbb{Q}^S}\left[\mathbf{1}_{S_t > K}\right] \\ &= S_0 \mathbb{Q}^S\left(S_t > K\right). \end{aligned}$$

Then, we simplify the expression found for V_0

$$V_0 = S_0 \mathbb{Q}^S \left(S_t > K\right) - K e^{-rT} \mathbb{Q}\left(S_t > K\right). \tag{3}$$

For the second term of (3), under the Black-Scholes model we have $S_t = S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right) T + \sigma W_T^{\mathbb{Q}}\right)$. We know that $S_T = S_0 e^Z$, where $Z \sim N\left(\left(r - \frac{\sigma^2}{2}\right) T, \sigma^2 T\right)$. Applying logarithms, we obtain that $\ln S_T = \ln S_0 + \left(r - \frac{\sigma^2}{2}\right) T + \sigma W_T^{\mathbb{Q}}$, and consequently

$$\ln S_T \sim \mathcal{N}\left(\ln S_0 + \left(r - \frac{\sigma^2}{2}\right) T, \sigma^2 T\right).$$

Then, note that we can write

$$\begin{aligned}
\mathbb{Q}\left(S_T > K\right) &= \mathbb{Q}\left(\ln S_T > \ln K\right) \\
&= \mathbb{Q}\left(\frac{\ln S_T - \ln S_0 - \left(r - \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}} > \frac{\ln K - \ln S_0 - \left(r - \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}}\right) \\
&= \mathbb{Q}\left(\frac{\ln S_T - \ln S_0 - \left(r - \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}} < \frac{-\ln K + \ln S_0 + \left(r - \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}}\right) \\
&= \mathbb{Q}\left(\frac{\ln S_T - \ln S_0 - \left(r - \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}} < \frac{\ln(S_0/K) + \left(r - \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}}\right) \\
&= \Phi(d_2),
\end{aligned}$$

where

$$d_2 = \frac{\ln(S_0/K) + \left(r - \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}}.$$

Note that in the third equality we have used the symmetry property $P(Z > z) = P(Z < -z)$.
For the first term of (3), considering that the drift is $(r + \sigma^2) S_t$ under $\mathbb{Q}^S$, we have that $S_t = S_0 \exp\left(\left(r + \sigma^2 - \frac{\sigma^2}{2}\right) T + \sigma W_T^{\mathbb{Q}^S}\right)$. We know that $S_T = S_0 e^Z$, where $Z \sim N\left(\left(r + \frac{\sigma^2}{2}\right) T, \sigma^2 T\right)$. Applying logarithms, we obtain that $\ln S_T = \ln S_0 + \left(r + \frac{\sigma^2}{2}\right) T + \sigma W_T^{\mathbb{Q}^S}$, and consequently

$$\ln S_T \sim \mathcal{N}\left(\ln S_0 + \left(r + \frac{\sigma^2}{2}\right) T, \sigma^2 T\right).$$

Then, note that we can write

$$
\begin{aligned}
\mathbb{Q}\left(S_T > K\right) &= \mathbb{Q}\left(\ln S_T > \ln K\right) \\
&= \mathbb{Q}\left(\frac{\ln S_T - \ln S_0 - \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} > \frac{\ln K - \ln S_0 - \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}\right) \\
&= \mathbb{Q}\left(\frac{\ln S_T - \ln S_0 - \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} < \frac{-\ln K + \ln S_0 + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}\right) \\
&= \mathbb{Q}\left(\frac{\ln S_T - \ln S_0 - \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} < \frac{\ln(S_0/K) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}\right) \\
&= \Phi(d_1),
\end{aligned}
$$

where

$$
d_1 = \frac{\ln(S_0/K) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}.
$$

Combining both terms, we obtain the Black-Scholes formula

$$
V_0 = S_0\mathbb{Q}^S(S_T > K) - Ke^{-rT}\mathbb{Q}(S_T > K) = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2).
$$

3 Two correlated stocks

Assume:

$$dS_1 = rS_1 dt + \sigma_1 S_1 dW_1^{\mathbb{Q}},$$
$$dS_2 = rS_2 dt + \sigma_2 S_2 dW_2^{\mathbb{Q}},$$

with

$$dW_1\, dW_2 = \rho\, dt.$$

(a) Derive the SDE of:

$$X_t = \frac{S_2(t)}{S_1(t)}.$$

We start by applying Ito's product rule to X_t, as S_1 and S_2 are both diffusion processes:

$$dX_t = \frac{1}{S_1(t)} dS_2(t) + d\left(\frac{1}{S_1(t)}\right) S_2(t) + dS_2(t) d\left(\frac{1}{S_1(t)}\right). \tag{4}$$

Now, we use Ito's lemma to find $d\left(\frac{1}{S_1(t)}\right)$:

$$\begin{aligned}
d\left(\frac{1}{S_1(t)}\right) &= \left(-\frac{1}{S_1(t)^2}\right) dS_1(t) + \frac{1}{2}\left(\frac{2}{S_1(t)^3}\right)(dS_1(t))^2 \\
&= -\frac{1}{S_1(t)^2} dS_1(t) + \frac{1}{S_1(t)^3}(dS_1(t))^2 \\
&= -\frac{1}{S_1(t)^2}\left(rS_1(t)dt + \sigma_1 S_1(t) dW_1^{\mathbb{Q}}\right) + \frac{1}{S_1(t)^3}\left(rS_1(t)dt + \sigma_1 S_1(t) dW_1^{\mathbb{Q}}\right)^2 \\
&= -\frac{1}{S_1(t)}\left(rdt + \sigma_1 dW_1^{\mathbb{Q}}\right) + \frac{1}{S_1(t)^3}\left(\sigma_1^2 S_1(t)^2 dt\right) \\
&= -\frac{1}{S_1(t)}\left(rdt + \sigma_1 dW_1^{\mathbb{Q}}\right) + \frac{1}{S_1(t)}\left(\sigma_1^2 dt\right) \\
&= \frac{1}{S_1(t)}\left(\sigma_1^2 - r\right) dt - \frac{\sigma_1}{S_1(t)} dW_1^{\mathbb{Q}},
\end{aligned}$$

where we have used in the first equality that $S_1(t)$ does not depend explicitly on time t. Now, we substitute the derivative of $\frac{1}{S_1(t)}$ into (4):

$$\begin{aligned}
dX_t &= \frac{1}{S_1(t)}\left(rS_2(t)dt + \sigma_2 S_2(t) dW_2^{\mathbb{Q}}\right) + \left(\frac{1}{S_1(t)}\left(\sigma_1^2 - r\right) dt - \frac{\sigma_1}{S_1(t)} dW_1^{\mathbb{Q}}\right) S_2(t) \\
&\quad + \left(rS_2(t)dt + \sigma_2 S_2(t) dW_2^{\mathbb{Q}}\right)\left(\frac{1}{S_1(t)}\left(\sigma_1^2 - r\right) dt - \frac{\sigma_1}{S_1(t)} dW_1^{\mathbb{Q}}\right) \\
&= rX_t dt + \sigma_2 X_t dW_2^{\mathbb{Q}} + \sigma_1^2 X_t dt - rX_t dt - \sigma_1 X_t dW_1^{\mathbb{Q}} - \rho\sigma_2\sigma_1 X_t dt \\
&= X_t\left(\left(\sigma_1^2 - \rho\sigma_1\sigma_2\right) dt - \sigma_1 dW_1^{\mathbb{Q}} + \sigma_2 dW_2^{\mathbb{Q}}\right).
\end{aligned}$$

Finally, we find that the SDE for X_t is

$$dX_t = X_t \left(\left(\sigma_1^2 - \rho\sigma_1\sigma_2 \right) dt - \sigma_1 dW_1^{\mathbb{Q}} + \sigma_2 dW_2^{\mathbb{Q}} \right). \tag{5}$$

(b) Choose S_1 as numéraire and define $\mathbb{Q}^{S_1}$:

$$\frac{d\mathbb{Q}^{S_1}}{d\mathbb{Q}} = \frac{S_1(T)/B_T}{S_1(0)/B_0}.$$

Derive:

- The new measure.
- The changes in dW_1 and dW_2.

Note that we can rewrite $d\mathbb{Q}^{S_1}/d\mathbb{Q}$ as

$$\begin{aligned}\frac{d\mathbb{Q}^{S_1}}{d\mathbb{Q}} &= \frac{S_1(T)/B_T}{S_1(0)/B_0} = \frac{e^{-rT} S_1(0) \exp\left(\left(r - \frac{\sigma_1^2}{2} \right) T + \sigma_1 W_1^{\mathbb{Q}}(T) \right)}{S_1(0)} \\ &= \exp\left(\sigma_1 W_1^{\mathbb{Q}}(T) - \frac{\sigma_1^2 T}{2} \right) = \exp\left(\int_0^T \sigma_1 dW_1(s) - \frac{1}{2} \int_0^T \sigma_1^2 ds \right).\end{aligned}$$

If we use this Radon-Nikodym derivative to change from a measure $\mathbb{Q}$ to a new measure $\mathbb{Q}^{S_1}$, Girsanov's theorem states that we can define a new standard Brownian motion, $W^{\mathbb{Q}^{S_1}}(t)$, under this new measure. Our parameter is $\theta(t) = \sigma_1$, then we can write

$$W_1^{\mathbb{Q}^{S_1}}(t) = W_1^{\mathbb{Q}}(t) - \sigma_1 t. \tag{6}$$

Therefore, the change in dW_1 can be written as

$$dW_1^{\mathbb{Q}^{S_1}} = dW_1^{\mathbb{Q}} - \sigma_1 dt.$$

In order to derive the change in $W_2^{\mathbb{Q}}$, we need to consider that $W_2^{\mathbb{Q}}$ and $W_1^{\mathbb{Q}}$ are correlated with $dW_1^{\mathbb{Q}} dW_2^{\mathbb{Q}} = \rho dt$. Then, using the Cholesky decomposition, we write

$$dW_2^{\mathbb{Q}} = \rho dW_1^{\mathbb{Q}} + \sqrt{1-\rho^2} dW^{\mathbb{Q}}, \tag{7}$$

where $W^{\mathbb{Q}}$ is independent from $W_1^{\mathbb{Q}}$ under $\mathbb{Q}$. As $d\mathbb{Q}^{S_1}/d\mathbb{Q}$ only depended on $W_1^{\mathbb{Q}}$, $W^{\mathbb{Q}}$ is unaffected by the measure change. So, $dW^{\mathbb{Q}} = dW^{\mathbb{Q}^{S_1}}$. Now, we substitute (6) in (7):

$$\begin{aligned}dW_2^{\mathbb{Q}} &= \rho \left(dW_1^{\mathbb{Q}^{S_1}} + \sigma_1 dt \right) + \sqrt{1-\rho^2} dW^{\mathbb{Q}^{S_1}} \\ &= \left(\rho dW_1^{\mathbb{Q}^{S_1}} + \sqrt{1-\rho^2} dW^{\mathbb{Q}^{S_1}} \right) + \rho\sigma_1 dt.\end{aligned}$$

Notice that the terms inside the brackets are just the Cholesky decomposition of $dW_2^{\mathbb{Q}^{S_1}}$. Then, we can write

$$dW_2^{\mathbb{Q}^{S_1}} = dW_2^{\mathbb{Q}} - \rho\sigma_1 dt.$$

Then, one can deduce that the new Brownian motion under $\mathbb{Q}^{S_1}$ is

$$W_2^{\mathbb{Q}^{S_1}} = W_2^{\mathbb{Q}} - \rho\sigma_1 t.$$

(c) Determine the new drifts and explain why the drift of X_t disappears.

In order to obtain the new drifts under the new measure $\mathbb{Q}^{S_1}$, we substitute $dW_1^{\mathbb{Q}^{S_1}}$ and $dW_2^{\mathbb{Q}^{S_1}}$ into the expressions for the diffusion processes S_1 and S_2. Let's start for the drift of S_1 under $\mathbb{Q}^{S_1}$:

$$\begin{aligned} dS_1 &= rS_1 dt + \sigma_1 S_1 dW_1^{\mathbb{Q}} = rS_1 dt + \sigma_1 S_1 \left(dW_1^{\mathbb{Q}^{S_1}} + \sigma_1 dt\right) \\ &= \left(r + \sigma_1^2\right) S_1 dt + \sigma_1 S_1 dW_1^{\mathbb{Q}^{S_1}}. \end{aligned}$$

Now, we compute the new drift of S_2 under $\mathbb{Q}^{S_1}$:

$$\begin{aligned} dS_2 &= rS_2 dt + \sigma_2 S_2 dW_2^{\mathbb{Q}} = rS_2 dt + \sigma_2 S_2 \left(dW_2^{\mathbb{Q}^{S_1}} + \rho\sigma_1 dt\right) \\ &= \left(r + \rho\sigma_1\sigma_2\right) S_2 dt + \sigma_2 S_2 dW_2^{\mathbb{Q}^{S_1}}. \end{aligned}$$

Finally, substituting $dW_1^{\mathbb{Q}^{S_1}}$ and $dW_2^{\mathbb{Q}^{S_1}}$ into (5), we get

$$\begin{aligned} dX_t &= X_t \left(\left(\sigma_1^2 - \rho\sigma_1\sigma_2\right) dt - \sigma_1 \left(dW_1^{\mathbb{Q}^{S_1}} + \sigma_1 dt\right) + \sigma_2 \left(dW_2^{\mathbb{Q}^{S_1}} + \rho\sigma_1 dt\right)\right) \\ &= X_t \left(\left(\sigma_1^2 - \rho\sigma_1\sigma_2 + \rho\sigma_1\sigma_2 - \sigma_1^2\right) dt + \sigma_2 dW_2^{\mathbb{Q}^{S_1}} - \sigma_1 dW_1^{\mathbb{Q}^{S_1}}\right) \\ &= X_t \left(\sigma_2 dW_2^{\mathbb{Q}^{S_1}} - \sigma_1 dW_1^{\mathbb{Q}^{S_1}}\right). \end{aligned}$$

The new drifts are $\left(r + \sigma_1^2\right) S_1$ for the SDE of the asset S_1, and $\left(r + \rho\sigma_1\sigma_2\right) S_2$ for the SDE of the asset S_2. Notice that we do not have drift term in X_t. This is a consequence of the fundamental theorem of asset pricing, which states that if there are no arbitrage opportunities in the market, then for any chosen numéraire N_t, there exists an equivalent martingale measure $\mathbb{Q}^N$ such that the price of any traded asset S_t, when expressed in units of that numéraire, is a martingale under $\mathbb{Q}^N$. In our case, the numéraire is S_1, and the e.m.m. is $\mathbb{Q}^{S_1}$. In consequence as $X_t = S_2(t)/S_1(t)$ (i.e. the price of asset 2 expressed in units of asset 1), then, X_t must be a martingale under $\mathbb{Q}^{S_1}$. This means that the drift of X_t must disappear, and only diffusion terms remain. If a drift term remained, it would be a predictable trend, violating the martingale property and implying an arbitrage opportunity.

4 Monte Carlo pricing of a European basket call

$$(A_T - K)^+, \qquad A_T = \frac{1}{2}(S_1(T) + S_2(T)).$$

Notation:

- M = number of simulated paths.
- $Y^{(m)}$ = discounted payoff for path m.
- $\widehat{V}(M)$ = Monte Carlo estimator.
- $\widehat{\mathrm{SE}}(M)$ = standard error.

Estimator:

$$\widehat{V}(M) = \frac{1}{M}\sum_{m=1}^{M} Y^{(m)}, \qquad \widehat{\mathrm{SE}}(M) = \frac{s(M)}{\sqrt{M}}.$$

Scenario 1

- Spots: $[100, 100]$
- Volatilities: $[20\%, 20\%]$
- $r = 6\%$
- $\rho = 0.9999$
- $T = 2$
- $K = 100$
- $M \in \{10k, 25k, 50k, 75k, 100k\}$

1. Simulate under $\mathbb{Q}$ and compute price and SE table.
2. Simulate under $\mathbb{Q}^{S_1}$ using dynamics from Question 3.

M	Q price	Q SE	QS1 price	QS1 SE
10,000	17.1699	0.23224	17.0899	0.16286
25,000	17.1125	0.14652	17.0799	0.10273
50,000	17.0974	0.10288	17.1274	0.07235
75,000	17.1464	0.08404	17.1611	0.05912
100,000	17.1849	0.07278	17.1943	0.05122

Figure 4: Prices and SE under $\mathbb{Q}$ and $\mathbb{Q}^{S_1}$ for different values of M.

3. Compare estimators and SEs; verify $1/\sqrt{M}$ behavior; explain variance reduction.

   Note that both estimators to approximately the same value (17.18 for the estimator under $\mathbb{Q}$ and 17.19 for the estimator under $\mathbb{Q}^{S_1}$) as the number of paths M increases. This shows the convergence of the Monte Carlo estimator to the expected value of the payoff. The small differences between estimates are due to sampling noise and they decrease with M.

   Also, we know that the SE satisfies $\hat{SE}(M) = \frac{s(M)}{\sqrt{M}}$. This can be observed in our numerical implementation: from $M = 10k$ to $M = 100k$, we multiply the number of paths by 10, and the SE decreases from 0.232 to 0.073 under $\mathbb{Q}$, which is a ratio of $0.073/0.232 \approx 0.313 \approx 1/\sqrt{10}$, which is consistent with the Central Limit Theorem (CLT) prediction. Under $\mathbb{Q}^{S_1}$, it decreases from 0.163 to 0.051, which is a ratio of $0.051/0.163 \approx 0.313 \approx 1/\sqrt{10}$, which is also consistent with the CLT.

   The $\mathbb{Q}^{S_1}$ estimator achieves a lower standard error than the $\mathbb{Q}$ estimator for all values of M. At $M = 100,000$, we have that $SE^{\mathbb{Q}^{S_1}} = 0.051 < 0.073 = SE^{\mathbb{Q}}$. This variance reduction is explained by the effective volatility of the ratio $X_t = S_2(t)/S_1(t)$ under $\mathbb{Q}^{S_1}$. From Exercise 3(c), follows $dX_t = X_t \left(\sigma_2 dW_2^{\mathbb{Q}^{S_1}} - \sigma_1 dW_1^{\mathbb{Q}^{S_1}}\right)$ under $\mathbb{Q}^{S_1}$. Computing the quadratic variation of the diffusion term using $dW_i^2 = dt$ and $dW_i dW_j = \rho dt$ gives the volatility:

$$\tilde{\sigma} = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} = \sqrt{0.04 + 0.04 - 2(0.9999)(0.04)} \approx 0.$$

   Since $\rho \approx 1$ and $\sigma_1 = \sigma_2$, the ratio $X_T = S_2(T)/S_1(T)$ is nearly constant under $\mathbb{Q}^{S_1}$. The payoff $(A_T - K)^+/S_1(T)$ therefore has very low variance, as almost all randomness cancels out between the two stocks. The variance of the $\mathbb{Q}^{S_1}$ estimator is dramatically lower, yielding much smaller SE for the same number of paths than the variance of the $\mathbb{Q}$ estimator.

4. Create a single plot:

   - x-axis: M
   - left y-axis: prices
   - right y-axis: SEs

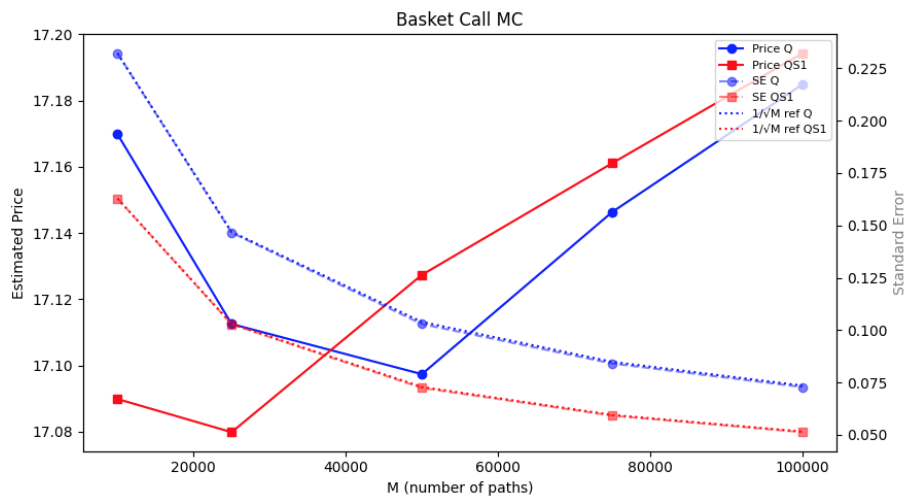

Figure 5: In the left y-axis, the estimated price for both $\mathbb{Q}$ and $\mathbb{Q}^{S_1}$ converging to the same value; In the right y-axis, the SE curves decaying along the $1/\sqrt{M}$ reference lines, with $\mathbb{Q}^{S_1}$ SE consistently below $\mathbb{Q}$ SE.

The $(1/\sqrt{M})$ reference curves are constructed by estimating the constant $(C = \widehat{\mathrm{SE}}(M_0) \cdot \sqrt{M_0})$ from the first data point $(M_0 = 10k)$, then plotting $(C/\sqrt{M})$. The agreement between the actual SE values and the reference curves confirms the $(1/\sqrt{M})$ decay predicted by the Central Limit Theorem.

Scenario 2

- Spots: $[443, 73]$
- Volatilities: $[10\%, 20\%]$
- $r = 6\%$
- $\rho = 0.2$
- $T = 2$
- $K = 100\% \times \frac{443+73}{2}$
- $M \in \{10k, 25k, 50k, 75k, 100k\}$

Repeat the above analysis.

1. Simulate under $\mathbb{Q}$ and compute price and SE table.

2. Simulate under $\mathbb{Q}^{S_1}$ using dynamics from Question 3.

M	Q price	Q SE	QS1 price	QS1 SE
10,000	32.3078	0.30529	32.3039	0.25325
25,000	32.4000	0.19384	32.3941	0.16072
50,000	32.4916	0.13628	32.5186	0.11321
75,000	32.4793	0.11117	32.5022	0.09234
100,000	32.5517	0.09651	32.5562	0.08017

Figure 6: Prices and SE under $\mathbb{Q}$ and $\mathbb{Q}^{S_1}$ for different values of M.

3. Compare estimators and SEs; verify $1/\sqrt{M}$ behavior; explain variance reduction.

   Note that both estimators to approximately the same value (32.55 for the estimator under $\mathbb{Q}$ and 32.56 for the estimator under $\mathbb{Q}^{S_1}$) as the number of paths M increases. This shows the convergence of the Monte Carlo estimator to the expected value of the payoff. The small differences between estimates are due to sampling noise and they decrease with M.

   Also, we know that the SE satisfies $\hat{SE}(M) = \frac{s(M)}{\sqrt{M}}$. This can be observed in our numerical implementation: from $M = 10k$ to $M = 100k$, we multiply the number of paths by 10, and the SE decreases from 0.305 to 0.097 under $\mathbb{Q}$, which is a ratio of $0.097/0.305 \approx 0.318 \approx 1/\sqrt{10}$, which is consistent with the Central Limit Theorem (CLT) prediction. Under $\mathbb{Q}^{S_1}$, it decreases from 0.253 to 0.080, which is a ratio of $0.080/0.253 \approx 0.316 \approx 1/\sqrt{10}$, which is also consistent with the CLT.

   The $\mathbb{Q}^{S_1}$ estimator achieves a lower standard error than the $\mathbb{Q}$ estimator for all values of M. At $M = 100,000$, we have that $SE^{\mathbb{Q}^{S_1}} = 0.080 < 0.097 = SE^{\mathbb{Q}}$. This variance reduction is explained by the effective volatility of the ratio $X_t = S_2(t)/S_1(t)$ under $\mathbb{Q}^{S_1}$. From Exercise 3(c), follows $dX_t = X_t \left(\sigma_2 dW_2^{\mathbb{Q}^{S_1}} - \sigma_1 dW_1^{\mathbb{Q}^{S_1}}\right)$ under $\mathbb{Q}^{S_1}$. Computing the quadratic variation of the diffusion term using $dW_i^2 = dt$ and $dW_i dW_j = \rho dt$ gives the volatility:

   $$\tilde{\sigma} = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} = \sqrt{0.01 + 0.04 - 2(0.2)(0.1)(0.2)} \approx \sqrt{0.042} \approx 20.5\%.$$

   Since $\rho \approx 0.2$ and we have distinct values for the volatilites($\sigma_1 = 10\%$ and $\sigma_2 = 20\%$), the ratio $X_T = S_2(T)/S_1(T)$ retains some volatility under $\mathbb{Q}^{S_1}$. The change of numéraire still reduces variance compared to the $\mathbb{Q}$ estimator, as it removes the drift term, but the reduction is less if we compare it with the one from Scenario 1.

4. Create a single plot:

- x-axis: M
- left y-axis: prices
- right y-axis: SEs

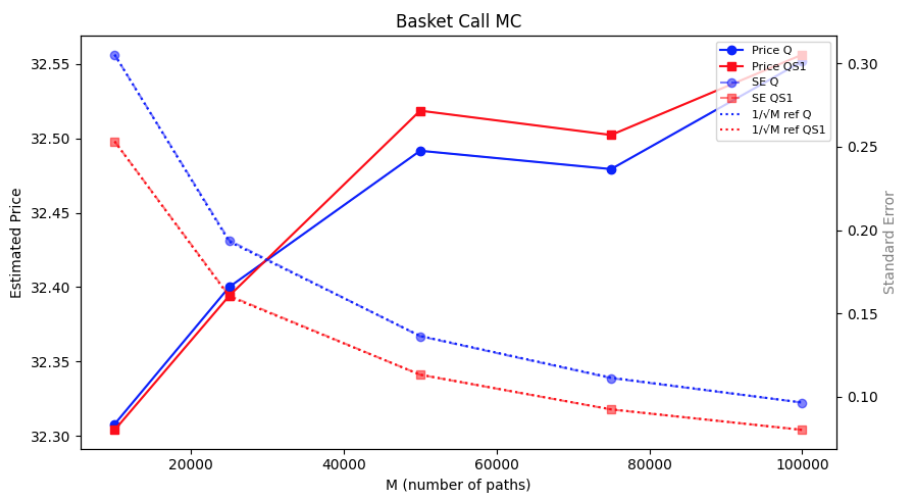

Figure 7: In the left y-axis, the estimated price for both $\mathbb{Q}$ and $\mathbb{Q}^{S_1}$ converging to the same value; In the right y-axis, the SE curves decaying along the $1/\sqrt{M}$ reference lines, with $\mathbb{Q}^{S_1}$ SE consistently below $\mathbb{Q}$ SE.

The $(1/\sqrt{M})$ reference curves are constructed by estimating the constant $(C = \widehat{\mathrm{SE}}(M_0) \cdot \sqrt{M_0})$ from the first data point $(M_0 = 10k)$, then plotting $(C/\sqrt{M})$. The agreement between the actual SE values and the reference curves confirms the $(1/\sqrt{M})$ decay predicted by the Central Limit Theorem.